

Question 1 (20marks)

- (a) Factor fully (i) $8x^3 - y^3$
(ii) $x^3 - x^2 - x + 1$
- (b) Simplify $\frac{c^2 + 5c + 6}{c^2 - 16} \div \frac{c + 3}{c - 4}$
- (c) Solve $\frac{3}{x-2} \leq 5$
- (d) i) Simplify $\frac{10\sqrt{21} \times \sqrt{22}}{\sqrt{55} \times 3\sqrt{35}}$
ii) Given that $a = 4 - \sqrt{15}$ simplify
 $\alpha) a + \frac{1}{a}$ $\beta) a^2 + \frac{1}{a^2}$
- (e) If $x + \sqrt{y-x} = 4 + \sqrt{5}$ find x and y
- (f) Given $f(x) = \frac{x^3 - 3x^2 + 1}{x(1-x)}$ for $x \neq 0, 1$ find $f\left(\frac{1}{x}\right)$ in simplest form

Question 2 (20 marks)

- (a) What is the natural domain of $y = \frac{1}{\sqrt{1-x^2}}$
- (b) If $f(x) = \frac{1-x}{x}$ find $f^{-1}(x)$
- (c) On separate axes sketch (i) $y = (x+1)^2 + 2$
(ii) $y = x(3-x)^2$
(iii) $y = \frac{1}{x(3-x)^2}$
- (d) Simplify (i) $\frac{|x|}{x}$ $x \neq 0$ (ii) $|2x-5| - |x-9|$ for $2\frac{1}{2} < x < 9$
- (e) By finding $f(-x)$ decide if the following functions are odd, even or neither
(i) $f(x) = \frac{a^2 - x^2}{a^2 + x^2}$ where a is a constant
(ii) $f(x) = |x^3 - x|$
- (f) By drawing appropriate graphs, or otherwise, solve (i) $|2x-6| \leq 4$ (ii) $x+6 > 2|x|$

END OF PAPER

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Year 11 Extension 1 2001

Question 1

$$(a) (i) \quad 8x^3 - y^3 = (2x)^3 - (y)^3 \\ = (2x - y)(4x^2 + 2xy + y^2)$$

$$(ii) \quad x^3 - x^2 - x + 1 = x^2(x - 1) - 1(x - 1) \\ = (x^2 - 1)(x - 1) \\ = (x - 1)(x + 1)(x - 1) \\ = (x + 1)(x - 1)^2$$

$$(b) \quad \frac{c^2 + 5c + 6}{c^2 - 16} = \frac{c + 3}{c - 4} \\ = \frac{(c + 3)(c + 2)}{(c - 4)(c + 4)} \times \frac{(c - 4)}{(c + 3)} \\ = \frac{(c + 2)}{(c + 4)}$$

$$(c) \quad \frac{3}{x - 2} \leq 5$$

$$(x - 2)^2 \times \frac{3}{(x - 2)} \leq 5(x - 2)^2$$

$$3(x - 2) \leq 5(x - 2)^2$$

$$3(x - 2) - 5(x - 2)^2 \leq 0$$

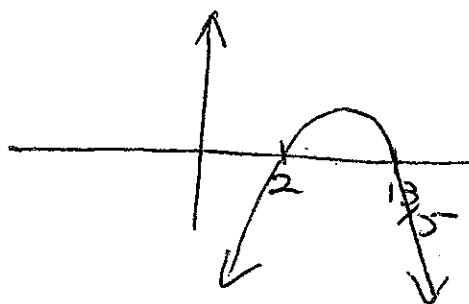
$$(x - 2)(3 - 5(x - 2)) \leq 0$$

$$(x - 2)(3 - 5x + 10) \leq 0$$

$$(x - 2)(13 - 5x) \leq 0$$

$$x = 2 \text{ or } \frac{13}{5}$$

$$\therefore x \leq 2 \text{ or } x \geq \frac{13}{5}$$



(2)

$$\begin{aligned}
 \text{(d) (i)} \quad \frac{10\sqrt{24} \times \sqrt{22}}{\sqrt{55} \times 3\sqrt{35}} &= \frac{10\cancel{\sqrt{7}} \times \sqrt{3} \times \sqrt{2} \times \cancel{\sqrt{11}}}{\sqrt{5} \times \cancel{\sqrt{11}} \times 3 \times \cancel{\sqrt{7}} \times \sqrt{5}} \\
 &= \frac{10\sqrt{6}}{15} \\
 &= \frac{2\sqrt{6}}{3}
 \end{aligned}$$

$$\text{(ii)} \quad a = 4 - \sqrt{15}$$

$$\alpha) \quad a + \frac{1}{a}$$

$$= \frac{4 - \sqrt{15}}{1} + \frac{1}{4 - \sqrt{15}}$$

$$= \frac{(4 - \sqrt{15})^2 + 1}{(4 - \sqrt{15})}$$

$$= \frac{16 - 8\sqrt{15} + 15 + 1}{(4 - \sqrt{15})}$$

$$= \frac{32 - 8\sqrt{15}}{4 - \sqrt{15}}$$

$$= \frac{8(4 - \sqrt{15})}{4 - \sqrt{15}}$$

$$= 8$$

$$\beta) \quad a^2 + \frac{1}{a^2}$$

$$= \left(a + \frac{1}{a}\right)^2 - 2$$

$$= (8)^2 - 2$$

$$= 64 - 2$$

$$= 62$$

$$\left(a + \frac{1}{a}\right)^2 = a^2 + \frac{1}{a^2} + 2$$

$$a^2 + \frac{1}{a^2} = \left(a + \frac{1}{a}\right)^2 - 2$$

(3)

Question 1 continued.

$$\Rightarrow \text{if } x + \sqrt{y-x} = 4 + \sqrt{5}$$

$$\therefore x = 4 \text{ and}$$

$$y - x = 5$$

$$\therefore y - 4 = 5$$

$$y = 9$$

$$\therefore x = 4 \text{ and } y = 9.$$

$$(f) \quad f(x) = \frac{x^3 - 3x^2 + 1}{x(1-x)} \quad x \neq 0, 1$$

$$f\left(\frac{1}{x}\right) = \frac{\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 + 1}{\frac{1}{x}\left(1 - \frac{1}{x}\right)}$$

$$= \frac{\frac{1}{x^3} - \frac{3}{x^2} + 1}{\frac{1}{x} - \frac{1}{x^2}} \times \frac{x^3}{x^3}$$

$$= \frac{x^3 \left(\frac{1}{x^3} - \frac{3}{x^2} + 1 \right)}{\frac{1}{x} - \frac{1}{x^2}}$$

$$= \frac{1 - 3x + x^3}{x^2 - x}$$

(4)

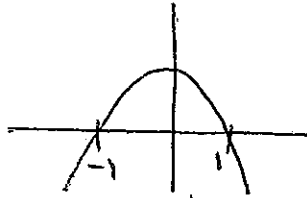
Question 2

(a) $y = \frac{1}{\sqrt{1-x^2}}$

$$\sqrt{1-x^2} > 0$$

$$1-x^2 > 0$$

$$(1-x)(1+x) > 0$$

 $\therefore$ Domain: $-1 < x < 1$

Q.2) If $y = \frac{1-x}{x} = f(x)$

Find $f^{-1}(x) \Rightarrow x = \frac{1-y}{y}$

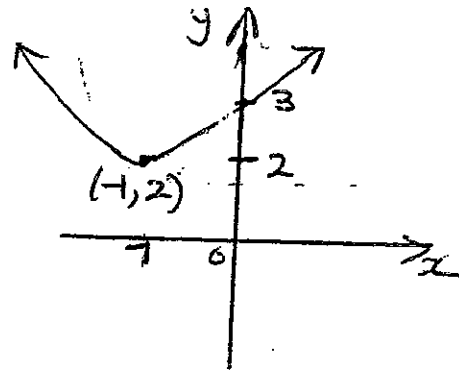
$$xy = 1-y$$

$$xy + y = 1$$

$$y(x+1) = 1$$

$$y = \frac{1}{x+1}$$

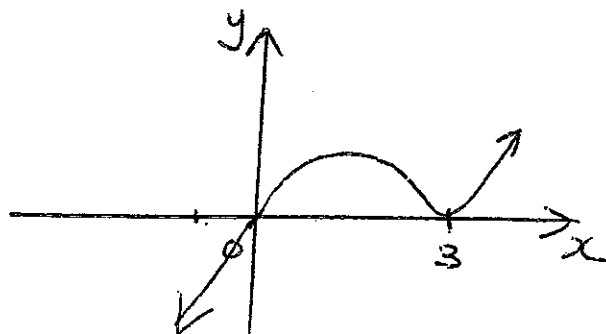
Q.2)(i) $y = (x+1)^2 + 2$



(ii) $y = x(3-x)^2$

$$x = 0, 3$$

x	-1	0	1	3	4
y	-16	0	4	0	4



$$y = \frac{1}{x(3-x)^2}$$

Domain: all real values, $x \neq 0$, $x \neq 3$

Odd/Even $f(x) = \frac{1}{x(3-x)^2}$

$$f(-x) = \frac{1}{(-x)(3-(-x))^2}$$

$$= \frac{1}{-x(3+x)^2}$$

$$= \frac{-1}{x(3+x)^2}$$

$$\neq f(x) = \frac{1}{x(3-x)^2}$$

$\therefore$ not odd nor even.

Intercepts $x \neq 0$ $\therefore$ no intercepts.
 $y \neq 0$

Vertical asymptotes

$$x=0, x=3$$

Horizontal asymptotes $x \rightarrow \infty$ $y \rightarrow ?$

$$\lim_{x \rightarrow \infty} \frac{1}{x(9-6x+x^2)}$$

$$\lim_{x \rightarrow \infty} \frac{1}{9x - 6x^2 + x^3}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x^3}}{\frac{9x}{x^3} - \frac{6x^2}{x^3} + \frac{x^3}{x^3}}$$

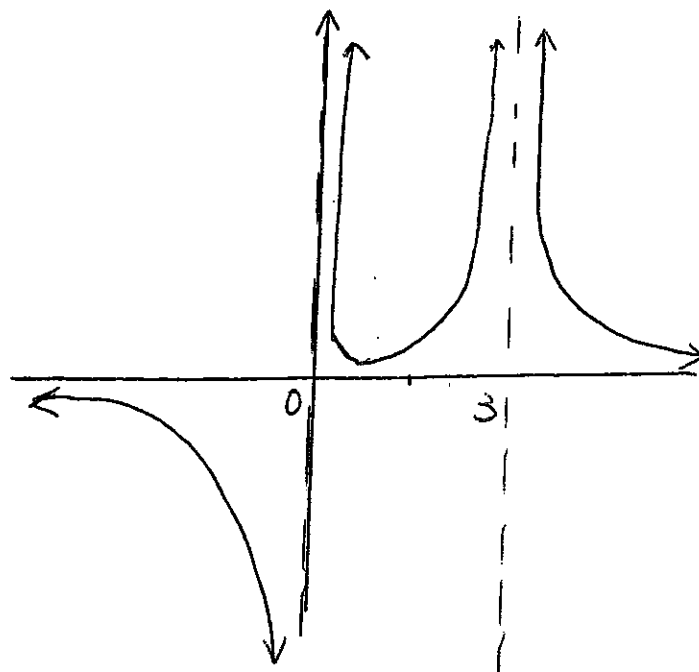
$$= 0$$

$$\text{As } x \rightarrow +\infty \quad y \rightarrow 0^+$$

$$\text{As } x \rightarrow -\infty \quad y \rightarrow 0^-$$

$$0 < x < 3$$

x	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	$\frac{1}{2}$
y	$\frac{1}{4}$	0.3	$\frac{1}{2}$	1.6	0.32



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Question 2 continued.

(c) (iii) $y = \frac{1}{x(3-x)^2}$

Using $y = x(3-x)^2$

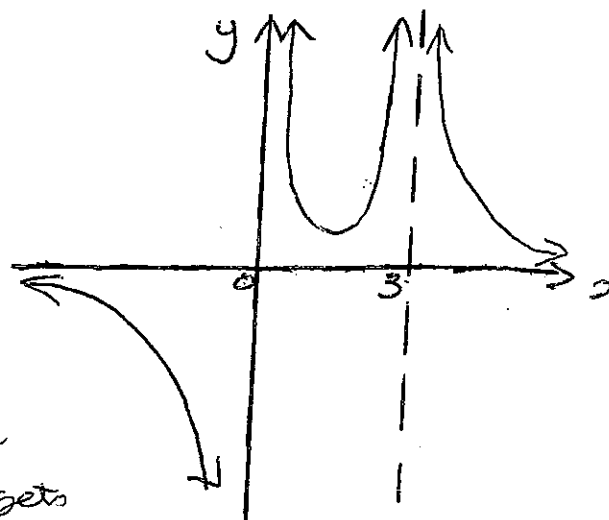
$y = 0$ at $x = 0$ and 3

$\therefore \frac{1}{y} \neq x = 0$ and 3 (asymptotes)

for $x \rightarrow 3$ $x \rightarrow \infty$ $\therefore$ for $\frac{1}{y}$, $x \rightarrow 0$ $\frac{1}{y} \rightarrow +\infty$

for $x < 0$ $x \rightarrow -\infty$ $\therefore$ for $\frac{1}{y}$ $x < 0$ $\frac{1}{y} \rightarrow -\infty$

for $0 < x < 3$, the largest point will become the minimum point. And as the function gets closer to zero $\frac{1}{y}$ will get closer to ∞ .



(c) $\frac{|x|}{x}$ $x \neq 0$

if $x < 0$

$|x| = -x$

$\therefore \frac{|x|}{x} = \frac{-x}{x}$
 $= -1$

if $x > 0$

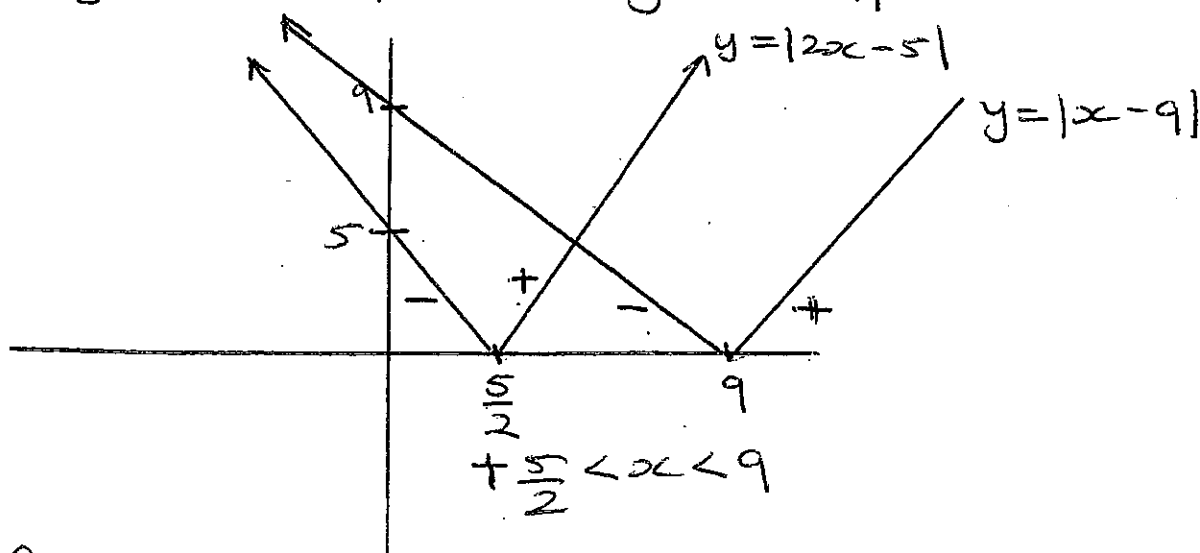
$|x| = +x$

$\frac{|x|}{x} = \frac{x}{x}$
 $= 1$

(ii) $|2x-5| - |x-9|$

sketch $y = |2x-5|$ and $y = |x-9|$

(c)



Simplify $|2x-5| - |x-9|$

for $+\frac{5}{2} < x < 9$
 $= +(2x-5) - -(x-9)$
 $= 2x-5+x-9$
 $= 3x-14$

(6)

Question 2 continued

$$(e) (i) f(x) = \frac{a^2 - x^2}{a^2 + x^2}$$

$$\begin{aligned} f(-x) &= \frac{a^2 - (-x)^2}{a^2 + (-x)^2} \\ &= \frac{a^2 - x^2}{a^2 + x^2} \end{aligned}$$

$$\therefore f(-x) = f(x)$$

$\therefore f(x)$ is an even function.

$$\begin{aligned} (ii) f(x) &= |x^3 - x| \\ f(-x) &= |(-x)^3 - (-x)| \\ &= |-x^3 + x| \\ &= |-1(x^3 - x)| \\ &= |-1| \cdot |x^3 - x| \\ &= 1 \cdot |x^3 - x| \\ &= f(x) \end{aligned}$$

$\therefore f(x)$ is an even function.

$$(f) (i) |2x - 6| \leq 4$$

$$2x - 6 \leq 4 \quad \text{or} \quad 2x - 6 \geq -4$$

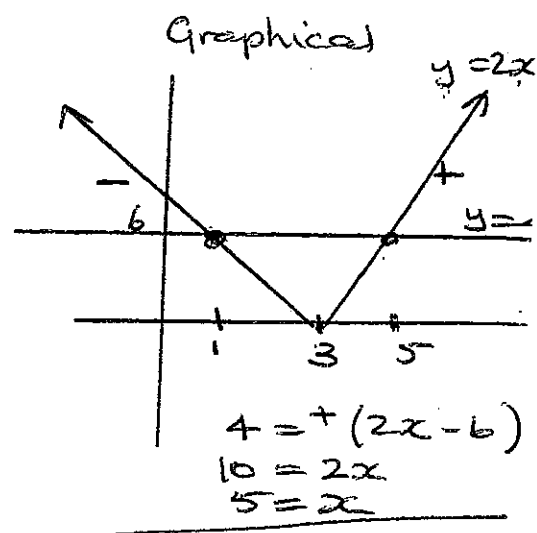
$$2x \leq 10 \quad \text{or} \quad 2x \geq 2$$

$$x \leq 5$$

$$x \geq 1$$



$$\therefore 1 \leq x \leq 5$$



$$\begin{aligned} 4 &= -(2x - 6) \\ 4 &= -2x + 6 \\ -2 &= -2x \\ 1 &= x \end{aligned}$$

$$-1 \leq x \leq 5$$